



Assign to class



Class Analytics



2.1 Parts of the Cell

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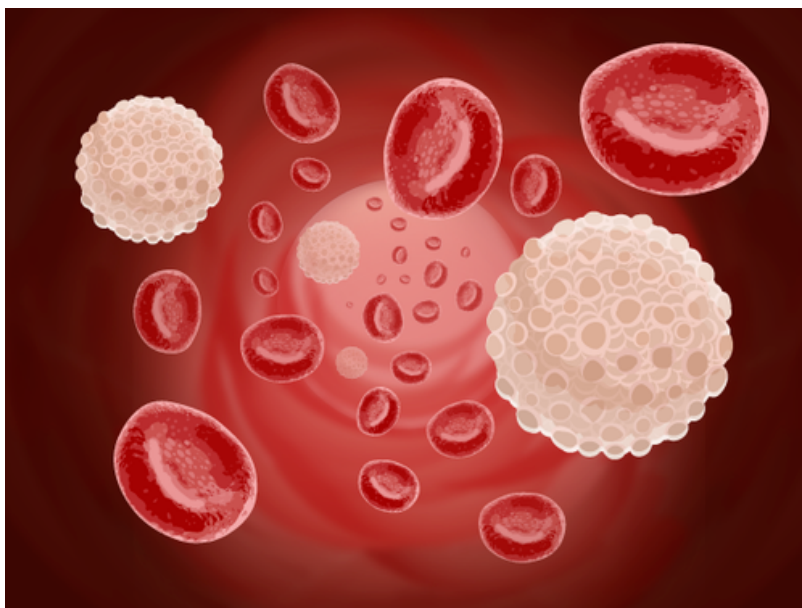
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What you will learn

- How and when **cells** were discovered
- The **cell theory**
- How diverse are cells
- What parts are common among all living cells



[Figure 1]

What do a bacterial cell and one of your cells have in common?

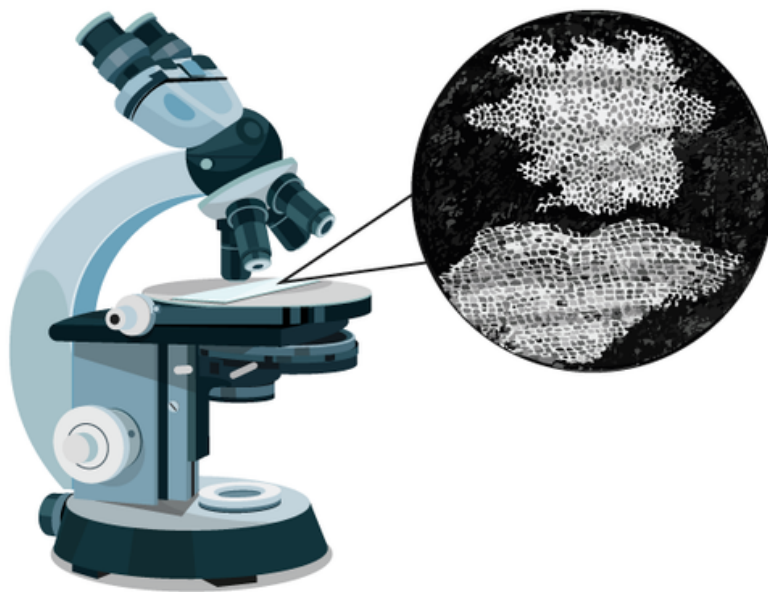


of the cell. Yet, all cells - cells from the smallest **bacteria** to those in the largest whale - do some similar functions, so they do have parts in common.

Common Parts of Cells

Discovery of Cells

The first time the word *cell* was used to refer to these tiny units of life was in 1665 by a British scientist named Robert Hooke. Hooke was one of the earliest scientists to study living things under a **microscope**. The microscopes of his day were not very strong, but Hooke was still able to make an important discovery. When he looked at a thin slice of cork under his microscope, he was surprised to see what looked like a honeycomb, made up of many tiny units, which Hooke called cells.



[Figure 2]

Illustration of cork cells. What type of material is cork? Do you know where cork comes from?

Soon after Robert Hooke discovered cells in cork, Anton van Leeuwenhoek in Holland made other important discoveries using a microscope. Leeuwenhoek made his own microscope lenses, and he was so good at it that his microscope was more powerful than other microscopes of his day. In fact, Leeuwenhoek's microscope was almost as strong as modern light microscopes.

Using his microscope, Leeuwenhoek discovered tiny **animals** such as rotifers. Leeuwenhoek also discovered human blood cells. He even scraped **plaque** from his own teeth and observed it under the microscope. What do you think Leeuwenhoek saw in the plaque? He saw tiny living things with a single cell that he named *animalcules* ("tiny animals"). Today, we call Leeuwenhoek's animalcules bacteria.

The Cell Theory

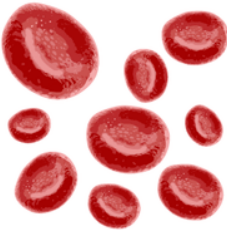
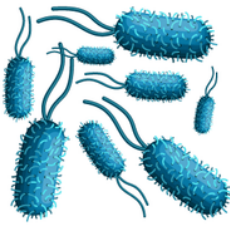
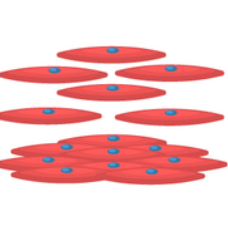
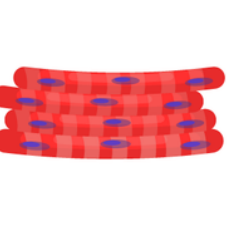
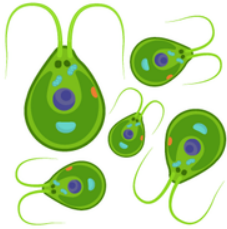
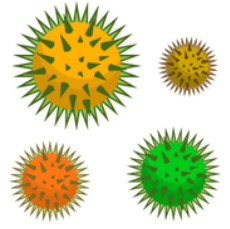
The cell theory is one of the fundamental **theories** of **biology**. For two centuries after the discovery of the microscope by Robert Hooke and Anton van Leeuwenhoek, biologists found cells everywhere. Biologists in the early part of the 19th century suggested that all living things were made of cells, but the role of cells as the primary building block of life was not discovered until 1839 when two German scientists, Theodor Schwann, a zoologist (studies animals), and Matthias Jakob Schleiden, a botanist (studies plants), suggested

cells. The collective **observations** of all three scientists form the cell theory, which states that:

- all **organisms** are made up of one or more cells,
- all the life functions of an organism occur within cells,
- all cells come from preexisting cells.

Cell Diversity

Cells with different functions often have different shapes. The cells pictured in the **Figure below** are just a few examples of the many different shapes that cells may have. Each type of cell in the figure has a shape that helps it do its job. For example, the job of the **nerve** cell is to carry messages to other cells. The nerve cell has many long extensions that reach out in all directions, allowing it to pass messages to many other cells at once. Do you see the tail-like projections on the **algae** cells? Algae live in **water**, and their tails help them swim. **Pollen** grains have spikes that help them stick to **insects** such as bees. How do you think the spikes help the pollen grains do their job? (*Hint: Insects pollinate **flowers**.*)

				
[Figure 3]	[Figure 4]	[Figure 5]	[Figure 6]	[Figure 7]
Red blood cells	Bacteria	Bone cell	Smooth muscle cells	Skeletal muscle cells
				
[Figure 8]	[Figure 9]	[Figure 10]	[Figure 11]	[Figure 12]
Algae	Pollen grains	Nerve cell	Columnar epithelial cells	Cardiac muscle cells

Common Parts of a Cell

Although cells are diverse, all cells have certain parts in common. The parts include a plasma membrane, **cytoplasm**, **ribosomes**, **cytoskeleton**, and **DNA**.

1. The **cell membrane** (also called the **plasma membrane**) is a thin coat of **lipids** that surrounds a cell. It forms the physical boundary between the cell and its environment, so you can think of it as the “skin” of the cell.
2. **Cytoplasm** refers to all of the cellular material inside the cell membrane, other than the **nucleus**. Cytoplasm is made up of a watery substance called **cytosol** and contains other cell structures such as ribosomes.
3. **Ribosomes** are structures in the cytoplasm where **proteins** are made.

The **cytoskeleton** consists of filaments and tubules that crisscross the cytoplasm and help maintain the cell's shape.

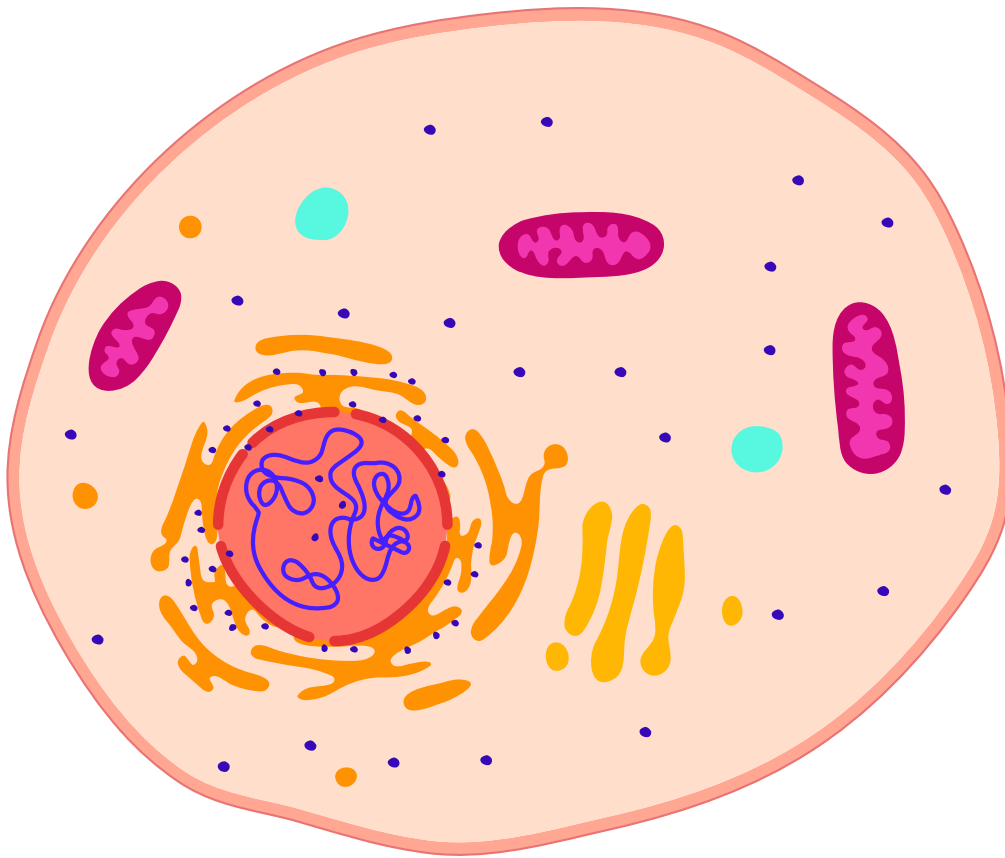
Cell Membrane

Cytoplasm

Ribosomes

Cytoskeleton

DNA



These parts are common to all cells, from organisms as different as bacteria and human beings. How did all known organisms come to have such similar cells? The similarities show that all life on Earth has a common evolutionary history.



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Summary

- The cell theory is one of the fundamental theories of biology. It states that all organisms are made up of one or more cells, all the life functions of an organism occur within cells, and all cells come from preexisting cells.
- Cells come in many different shapes. Cells with different functions often have different shapes.
- Although cells come in diverse shapes, all cells have certain parts in common. These parts include the cell membrane, cytoplasm, ribosomes, cytoskeleton, and DNA.

Review

1. Who coined the term *cell*, in reference to the tiny structures seen in living organisms?
2. Who identified *animalcules*? What are *animalcules*?
3. What are the three main parts of the cell theory?
4. List the four parts common to all cells.
5. What are the cell structures where proteins are made?
6. What is the role of DNA?



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